INKLOOM AI HACKATHON • CODENEXUS INDIA 2026 • ROUND 2: THE BUILD

Brand Rule Lab

Pre-publication review with a verifiable claim-to-fact ledger

An intelligent brand governance platform built for high-scrutiny marketing. It catches off-brand tone, wrong terminology, unsupported claims, and inaccessible visuals before publication. Every approved claim receives an immutable hash and stable identifier directly tied to the exact approved fact version that substantiates it.

Team: rohitlongesha5 (Rohit Longesha) **Focus Market:** India D2C Personal Care & Beauty **Status:** 216 Tests Passed (100% Green) • Live NVIDIA NIM & Gemini AI

1 DRAFT → 2 CHECK → 3 SUGGEST → 4 APPROVE → 5 LEDGER → 6 MONITOR

The Real Industry Problem

ASCI SCRUTINY

In fast-moving digital marketing, rapid-result claims and absolute promises face strict scrutiny from regulatory bodies and consumers:

90%	82%	1–4 d
ASCI CASES NEEDED EDITS	SCRUTINISED IN DIGITAL ADS	PEER TURNAROUND DELAY

> *"Annoying part design ka approval ni milta... senior deny krdete late factual edit pe."* — Peer interview evidence.

Sequential sign-offs, subjective comments ("make it pop"), and unrecorded guideline preferences lead to costly re-works and delayed campaigns.

Our Wedge: The Claim Ledger

CORE WEDGE

Generic brand assistants offer text review without historical lineage. **Brand Rule Lab** links every approved claim to a verifiable evidence version:

- **Stable Identifier:** Each approved claim receives a unique 12-char hex `stable_claim_id` and SHA-256 normalized text hash.
- **Fact-Version Linkage:** Bound to a specific `fact_version_id` (e.g. Clinical Study F-112 v3).
- **Instant Drift Propagation:** Fact updates or expiries flag dependent claims in **1 SQL update with 0 LLM calls**.

> **Strict Boundary:** `UNLINKED`, `NOT_OBSERVED`, and `REVIEW_REQUIRED` remain distinct states. A missing match is never quietly treated as compliant.

External live edits require a pasted-text hash check or manual import to verify coverage.

Deterministic Drift Cascades (Zero-LLM Overhead)

SQL CASCADE

```sql
-- Fact Owner approves version 4 of F-112
UPDATE approved_claims
SET drift_status = 'REVIEW_REQUIRED',
    flagged_at = CURRENT_TIMESTAMP
WHERE fact_id = 'F-112' AND fact_version_id < 4;
-- Expiry sweep flags expired versions instantly
```

Why this matters:

- **0 LLM API calls:** Drift tracking runs entirely in the database engine.
- **0 Web crawls:** Immediate notification without scraping external sites.
- **1 Atomic SQL transaction:** Updates all registered dependencies without race conditions.

Three Role-Gated Views

SIMULATED ROLES

Role	Primary Workflow	Key Permissions
Creator	Author & revise marketing copy	Uploads drafts (PDF, DOCX, TXT, images); accepts/edits suggestions.
Reviewer	Approval queue & guideline audits	Audits findings; classifies rules; approves/rejects submitted claims.
Fact Owner	Evidence & regulatory specifications	Approves facts, sets validity periods & markets; bumps versions.

Python Enforces. LLM Language.

DETERMINISTIC CORE

Component	Responsible Agent	Principle
Exact Rules	Python Predicates	100% reproducible exact pass/fail; zero hallucinations.
Claim Extraction	Python + NIM / Gemini	Python slices spans & offsets; AI selects IDs only.
Fact Matching	SQLite Filter + AI	SQL filters by product/market; AI ranks relationship.
Tone & Advice	Advisory AI (Nemotron)	Contextual suggestions only. AI never auto-publishes.

Rule Readiness Audit System

SOURCE-VERIFIED GOVERNANCE

When brand guideline documents (PDF/DOCX/TXT/XLSX) are onboarded, extracted proposals undergo an audit gate before any rule enters the enforcement engine. Every active rule retains source quote and page-level provenance:

1. CODE_CHECKABLE (Active Rules)

Strictly mapped to supported Python predicates (`term_list` , `regex` , `numeric_range` , `date_window` , `required_text` , `wcag_contrast` , `asset_hash`). Must pass automated positive/negative self-tests and human reviewer sign-off.

2. AI_ASSISTED (Advisory Guidance)

Contextual guidance such as brand voice nuances and narrative style. Surfaces advisory suggestions during review; never compiled into blocking deterministic predicates.

3. HUMAN_JUDGEMENT (Manual Routing)

Subjective visual balance, aesthetics, and cultural nuances that cannot be decided algorithmically. Routed straight to human reviewer queues.

4. UNCLEAR (Clarification Required)

Ambiguous or contradictory source guideline wording. Explicitly held and blocked from enforcement until the brand owner logs a written clarification.

Responsible Design, Privacy & Prompt Safety

SAFETY BY DESIGN

- **Human Authority:** AI never approves claims or auto-publishes content. Reviewers and Fact Owners retain exclusive authority.
- **Prompt Injection Defense:** Draft text is treated as untrusted data. AI has zero tool/database-write capabilities. Deterministic Python rules run independently and cannot be suppressed by draft instructions.
- **Privacy & Data Minimization:** Uploaded binary files (PDF/DOCX/PNG) are parsed locally on the host. Raw files are never transmitted to external AI.
- **Dual AI Provider Resiliency:** Primary NVIDIA NIM (Nemotron-3-Nano-Omni) with automatic graceful fallback to Google Gemini (3.8-Flash).

Live Checks in Action (Synthetic Demonstration Suite)

MONSOON BEAUTY CO.

Check Domain	Draft Signal / Violation	System Enforcement & Finding	Intent-Preserving Suggestion
Tone (V-04)	*"Say goodbye to acne forever!"*	FAIL Absolute promise conflicts with approved calm voice rule V-04.	Suggest qualified wording: "*Helps clear blemishes*"; reviewer confirms.
Terminology (T-12)	*"Our 2.5% BPO face wash"*	FAIL Rule T-12 disallows abbreviation "BPO" in consumer-facing copy.	One-click intent-preserving replacement with approved INCI term: "*Benzoyl Peroxide*".
Claim vs. Fact (F-112)	*"Clears breakouts in 3 days"*	UNLINKED Approved clinical trial MS-2024-F112 observes 4-week timeline, not 3 days.	Suggest: "*Within 4 weeks of twice-daily use*" citing Study MS-2024-F112.
Offer Date Window	*"Diwali sale ends 15 Nov"*	FAIL `date_window` predicate check fails; approved brief campaign expires 12 Nov.	Flag for campaign owner: renew approved fact or update copy date.
Visual & Contrast	Banner graphic contrast ratio 2.5:1	FAIL Fails WCAG AA 4.5:1 threshold for normal text legibility.	Change text foreground/background color pair to achieve ≥ 4.5:1.
Asset Integrity	Uploaded product image altered	FAIL Exact SHA-256 identity does not match approved product asset record.	Re-select approved asset from registry or submit for Fact Owner review.

Technology & Architecture Disclosures

PRODUCTION READY

- **UI:** Streamlit 1.66.1 (multi-role views, reactive diff & audit inspectors).
- **Primary AI:** NVIDIA NIM HTTPS API (`nemotron-3-nano-omni-30b-a3b-reasoning`).
- **Fallback AI:** Google Gemini API (`gemini-3.8-flash`) with auto-switchover.
- **Local Parsing:** PyMuPDF (PDF), python-docx (DOCX), openpyxl (XLSX), Tesseract OCR (images).
- **Database:** SQLite 3 with WAL mode, foreign keys, and additive schema migrations.
- **Contracts:** jsonschema 4.23.0 enforcement on all LLM responses.
- **Testing:** 216 pytest unit/integration tests (**216 passed, 0 failed**).

Round 3 Future Roadmap (Next Steps)

FUTURE SCOPE

1. Pre-Flight CMS & Social Gatekeeper: Direct webhooks for Shopify, WordPress, and Hootsuite that intercept publication if `stable_claim_id` is missing or drifted.

2. Multimodal Packaging & Video Frame Inspection: Temporal analysis using Whisper for audio disclaimers and frame extraction for video overlays and packaging renders.

3. Enterprise RBAC & Cryptographic Merkle Export: Single Sign-On (OIDC/SAML) and mathematically tamper-evident Merkle-tree claim exports for ASCI/FTC legal compliance audits.

Deployment & GitHub Repository Structure

READY TO DEPLOY

The project repository includes a production Dockerfile with Tesseract OCR dependencies, automated database migrations, and healthcheck probe. All 66 source files are tracked in Git on branch `main`, with local credentials (`.env`) and virtual environments (`.venv/`) completely excluded.